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Machine Learning and Deep Learning (Deep Learning)

An Explainable Deep Learning Framework for Breast
Cancer Classification from Histopathological Images
Using CNNs and Vision Transformers.

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1 MOTIVATION AND RATIONALE

Breast cancer is one of the most prevalent and deadly illnesses of women in the world. Early and timely diagnosis is a key factor in increasing survival rates and treating less complicated. This intractable manual diagnosis is, however, time-consuming and laborious and relies very much on the expertise of pathologists, with a high risk of misdiagnosis. The emergence of digital pathology has allowed the development of large facial-histopathological image datasets like BreakHis and make a development of machine learning and deep learning techniques possible to automatically classify the dataset in cancerous and non-cancerous images. These techniques have been found to be highly effective in aiding doctors with their diagnosis by enhancing accuracy and productivity. Though tremendous advancement has occurred in the field of breast cancer image classification, the field is still confronted with a number of problems. This task is very complicated due to the differences in tissue structure, staining and the high degree of intra-class similarity of both benign and malignant samples. Moreover, most of the already developed models are, unfortunately, not interpretable, and in medical applications, transparency in decision-making is needed. Therefore, to address these challenges motivating this study, the framework of comparative deep learning for the classification of breast cancer by histopathological image is proposed. It also compares several cutting-edge architectures such as CNN-based models and Vision Transformers to find the most effective model to accurately classify. Moreover, explainable AI methods like Grad-CAM++ are added to explain the interest areas in the histopathology pictures, adding to the intuitiveness and legitimacy of clinical choices.

The goal of this work is to help create a reliable and efficient computer aided diagnostic system that will assist Pathologists in identifying early stages of breast cancer and therefore be able to increase patient survival.

2 STATE OF THE ART (SOTA)

In recent years, tremendous progress has been achieved in breast cancer diagnosis based on histopathological images, which was initiated by the deep learning techniques. The handcrafted features were used for the breast cancer classification using the traditional machine learning classifiers like Support Vector Machines (SVM), K-Nearest Neighbors (KNN) and Random Forest (RF) classifiers. However, little of the complex spatial pattern has been captured from the images from histopathology to other approaches only to a limited extent in breast cancer diagnosis [1]. Deep learning has become the most prevalent solution for medical image analysis problems, due to Convolutional Neural Network (CNNs) being applied to these problems with high frequency. Due to their success in the field medical image analysis, Convolutional Neural Networks (CNNs) have become the most common solution for these problems. Architectures such as AlexNet, VGGNet, ResNet and DenseNet have yielded better results with the automated breast cancer classification tasks. Among them, ReGetNet is the most successful for this task of reusing features, particularly well suited for medical imaging tasks, as noted in gitereis [2]. Recently, the architectures of iEfficientNet and Xception have been introduced, these architectures are efficient architecture that was improved due to the high accuracy with low computational costs. These models use optimized Convolutional structures for further fine-grained tissue pattern in histopathological images. In the meantime, a new paradigm called Vision Transformers (ViT) has been presented in the medical image classification context. Currently, another variant determined by Vision Transformers (ViT) is suggested as an alternative paradigm in medical image classification. Unlike CNNs, ViT models leverage their self-attention properties for capturing global dependencies, allows models to compete effectively with the CNN in large-scale datasets [3]. Although some deep learning models are successful, their interpretability is lacking especially in clinical applications. Explainable AI (XAI) methods like Grad-CAM and Grad-CAM++ aid in visualizing model attention areas, boosting transparency and creating trust within the prediction process.

Although significant progress has been made in breast cancer classification on the BreakHis

dataset is still very challenging due to the lack of balance among classes, the variation from one patient to another and the differences between benign and malignant tissues. Hence, testing of deep learning and transformer enabled models is a must to determine strong architectures for accurate diagnosis.

3 OBJECTIVES

The main goal of this research is to build an interpretable and a valid deep learning system for classification of breast cancer, based on histopathological images. The particular aims of this research are given below:

- **Designing an Automated Classification System** – To design a deep learning framework which is able to differentiate between benign and malignant breast cancer histopathology images.
- **Compare State-of-the-Art Models** – to compare many of these architectures including CNN-based models and Vision Transformers like DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3 and ViT-B/16.
- **Enhance With Transfer Learning and Fine-tune On BreakHis** – To better classify the BreakHis dataset by fine tuning and overall transfer learning.
- **Enable Model Interpretability** – To enable methods known as explainable AI (Grad CAM++) to visualize the regions of interest (ROI) in histopathological images.
- **Minimise False Negatives** – To maximise recall in model evaluation, meaning to not misclassify any negative cases (malignancy), failures should be minimised.
- **Support Clinical Decision Making** – To help healthcare workers capture accurate cancer diagnostics predictions, which are interpretable by them.
- **Set up a Benchmark Comparison** – to have a full comparison of the performance of deep learning and transformer-based models as a future benchmarking.

4 METHODOLOGY

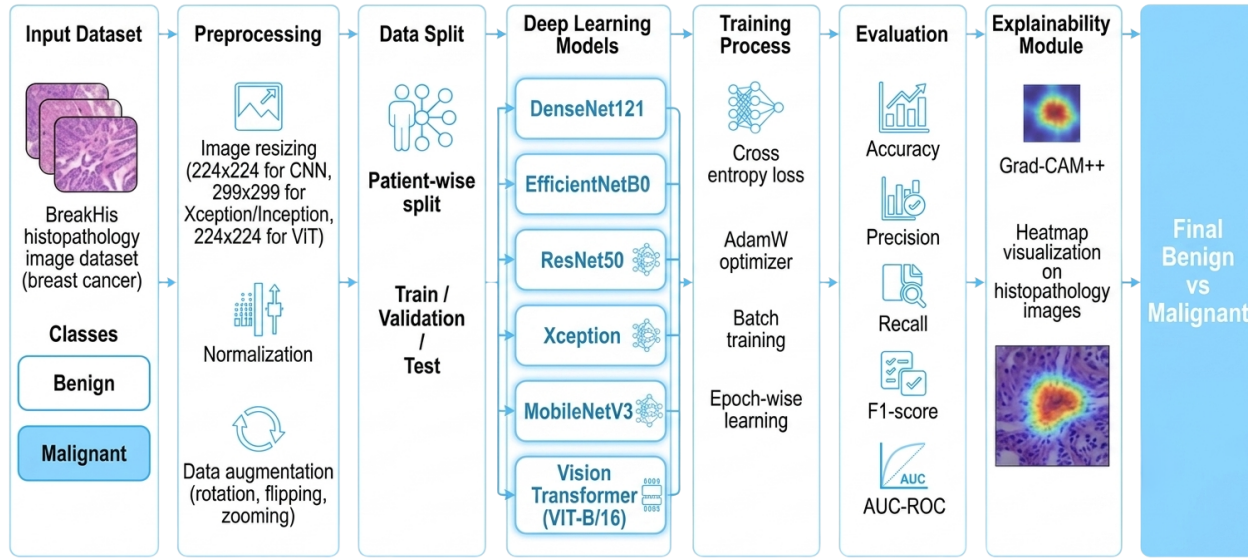


Figure 4.1: Proposed methodology for breast cancer classification using deep learning models and explainable AI.

The figure 4.1 presents a comparative study between deep learning models and Vision Transformer (ViT) models applied to the classification of breast cancer images to human histopathological image dataset BreakHis. The first step involves the images being resized, normalized, and subjected to various data augmentation methods like rotation, flipping, and zooming to make the model robust against transformations. To avoid the bias of evaluation, the dataset is split into a patient-wise training, validation and test split. The models trained and compared are DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3 and Vision Transformer (ViT-B/16). The training is implemented with cross entropy loss function, AdamW optimizer, batch training and epoch-wise learning. The metrics for modelling results are: Accuracy, Precision, Recall, F1-score and AUC-ROC. Grad-CAM++ is carried out to make the results easier to interpret and create heatmaps that determine regions of the image that all or a portion of the model relied on for prediction. Lastly, the most successful model is chosen to differentiate between two types of breast cancer (benign and

malignant), with Xception turning out to be the most successful one as per this study.

4.1 Dataset Collection

This study is based on the BreakHis dataset which includes histopathological images from breast tumour tissues. The data set is divided into either of the classes, “benign” or “malignant.” They take photographs of the image at different magnification levels, and train the model with those images they captured and test it on the ones they didn’t. At different magnifications, the images are taken at different Zoom levels and they use the images to train and test their model at different magnifications [4].

4.2 Data Preprocessing and Splitting

The size of all histopathological images have been fixed as per deep learning architectures requirement. CNN-based models used the 224×224 image size, while Xception and Inception models generated at 299×299 size. To facilitate the training of the model with sufficient stability and efficiency, Pixel values were normalised with the mean and standard deviation of the ImageNet set. Random rotations, horizontal and vertical flipping of images, and zooming were used to augment the data and reduce the possibility of over-fitting and increase model generalization. The training, validation and test sets were made with a ratio of 70:15:15 with the same class balance within each set. This splitting strategy is a stratified one to avoid data leakage and helps to fairly and reliably assess model performance.

4.3 Model Architectures

With this study, various types of deep learning architectures are tested to see and contrast their performance on breast cancer classification [5]. These include:

- **DenseNet121:** DenseNet121 is a dense CNN with dense connections from one layer to another, which allows for the reuse of features and can facilitate the smooth flow of information

from one layer to another. It is an architecture that minimizes the parameters, enhances feature propagation and classification performance.

- **EfficientNetB0:** EfficientNetB0 is a lightweight and efficient CNN, which scales up network depth, width and resolution in a balanced manner. It performs well with less number of parameters than conventional CNN architectures.
- **ResNet50:** ResNet50 is a Deep Residual Neural Network that has 50 layers. It helps to resolve the vanishing gradient problem with residual connections: it is a way of making deeper and better networks.
- **Xception:** Xception is an advanced Convolutional Neural Network (CNN) that is designed using a technique called depthwise separable convolutions. It enhances the abilities of feature extraction and has shown excellent performance on several medical image classification problems.
- **MobileNetV3:** MobileNetV3 is a recently published lightweight neural network for low memory devices. It incorporates squeeze and excitation modules and depthwise separated convolutions along with the neural architecture search to deliver high accuracy at low computation.
- **Vision Transformer (ViT-B/16):** Vision Transformer (ViT-B/16) extends the use of the transformer model to the domain of image classification, segmenting images into fixed-size patches, with the help of self-attention mechanisms. It provides a good grasp of the image relationship at the local and global level.

Pre-trained images of ImageNet are used as the initial conditions for each model and fine-tuned on the BreakHis database.

4.4 Training Strategy

Cross-Entropy Loss function is used for all models and AdamW optimizer for optimization. Weight decay is used for regularization and the learning rate is $1e-4$ [6], [7]. Optimizing is optimised by training several epochs, expanding on individual training batches.

To keep the best model according to the validation result, early stopping and model checkpointing are adopted.

4.5 Evaluation Metrics

Accuracies, precisions, recalls, F1-scores and receiver operating characteristic curve Area Under the Curve (AUC) were used to measure the performance of the proposed models. These metrics are based on True Positive (TP), True negative (TN), False positive (FP) and False negative (FN) are components of confusion matrix.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.3)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.4)$$

$$\text{AUC} = \int_0^1 TPR(FPR) d(FPR) \quad (4.5)$$

The accuracy is defined as the percentage of correctly classified samples. Precision measures the percentage of positive responses that are actually positive. In medical diagnosis, it is important that the model correctly identify cases when they are indeed positive, hence sensitivity is important. F1 Score is a harmonic average of Precision and Recall, it is used to measure the performance of classification. Finally, the Area Under the Receiver Operating Characteristic Curve (AUC) measures the model's accuracy or discriminative ability over various decision thresholds, and a higher AUC means the model is more accurate.

4.6 Explainability Using Grad-CAM++

Based on the aim of interpretability of models, Grad-CAM++ is used to visualize the region of the histopathological images that contribute most towards the model's predictions. This becomes useful in comprehending decisions made by the model and to ensure trust in the clinical use [8].

Heat-maps generated highlight important tissue structure and regions of abnormality that can affect the classification outcome.

5 EXPERIMENTS AND RESULTS

The setup of experiments, performance comparison and measurement of various deep learning and transformer models for the classification of breast cancer images from the BreakHis histopathological image database are described. The experimental setup and performance comparison/measurement of multiple deep learning and transformer-based models in breast cancer classification using the BreakHis histopathologic image database are provided. Here are the models that have been evaluated: DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3, and Vision Transformer (ViT-B/16).

5.1 Experimental Setup

Experiments were all carried out on a Python and GPUs programming interface called PyTorch. The weights for all Models were frozen using the pre-trained ImageNet weights and fine-tuned the model on the BreakHis dataset.

Table 5.1: Training Configuration

Parameter	Value
Loss Function	Cross-Entropy Loss
Optimizer	AdamW
Learning Rate	1×10^{-4}
Batch Size	16 or 32
Epochs	10
Image Size (CNNs)	224×224
Image Size (Xception)	299×299

Table 5.1 summarizes the hyperparameters and training settings used for all deep learning models in this study.

5.2 Evaluation Metrics

All models were tested with regard to Accuracy, Precision, Recall, F1-score, and Area Under the curve (AUC). These metrics enable a full measurement particularly for medical classification applications which have a high dependence on the absence of a false negativ

5.3 Comparative Results of Models

The results of the experiment are presented in column 2 to compare the different architectures on the BreakHis dataset. The performance summary is showed as follows:

Table 5.2: Performance Comparison of Deep Learning Models on the BreakHis Dataset

Model	Accuracy	Precision	Recall	F1-score	AUC
ViT-B/16	0.8157	0.8270	0.9035	0.8636	0.8839
MobileNetV3	0.8182	0.8465	0.8774	0.8617	0.8500
ResNet50	0.8148	0.8423	0.8774	0.8595	0.8855
Xception	0.7938	0.8107	0.8879	0.8475	0.8446
EfficientNetB0	0.7828	0.8301	0.8344	0.8322	0.8503
DenseNet121	0.7618	0.8056	0.8318	0.8185	0.8404

The work’s results comparing the performances of the six deep learning models provided at the breast cancer dataset, namely BreakHis, are summarized in Table 5.2. For all the models evaluated, the highest F1 score was 0.8636 with the best recall was 0.9035 for ViT-B/16 model, showing an excellent performance in recognising malignant cases. In terms of accuracy and precision results, the best performance was achieved with MobileNetV3, which reached an accuracy of 0.8182 and a precision of 0.8465, demonstrating a good balance between the predictive ability and resource efficiency of the system. The ResNet50 model obtained the best two classification performances with the highest AUC (0.8855) for the different classification thresholds.

In analysis, Xception obtained a high value for recall (0.8879) and F1-score (0.8475) with competitive result whereas the EfficientNetB0 model according to analysis had a moderate value for all evaluation metrics. For all overall results, DenseNet121 resulted in having the least overall

performance of the CNN models. In summary, ViT-B/16, MobileNetV3 and ResNet50 proved to be the best models overall, with each one outperforming the others in one area.

5.4 Analysis of Results

5.4.1 DenseNet121

DenseNet121 achieved an accuracy of 76.18%, precision of 80.56%, recall of 83.18%, F1-score of 81.85%, and AUC of 84.04%. Despite its poor overall performance, DenseNet121 showed good performance for classification due to the dense connectivity architecture which enables efficient reuse of features.

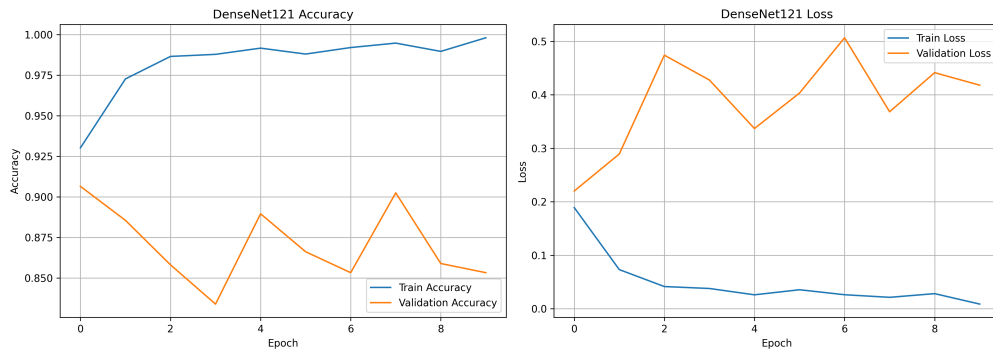


Figure 5.1: Training and Validation Accuracy/Loss Curves of DenseNet121

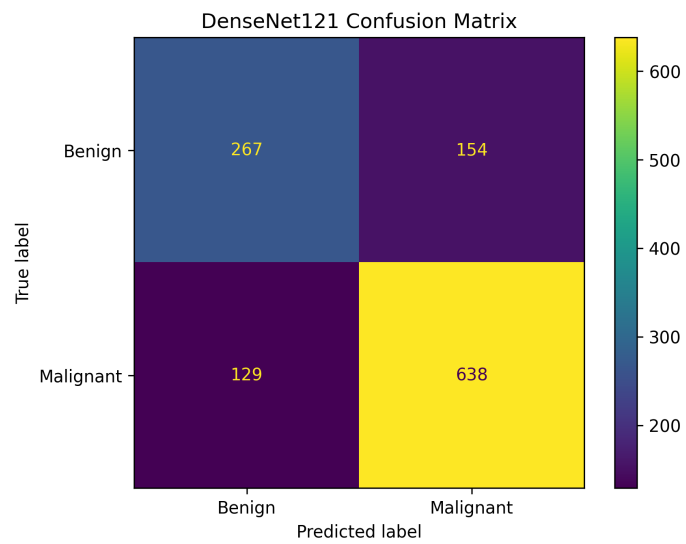


Figure 5.2: Confusion Matrix of DenseNet121

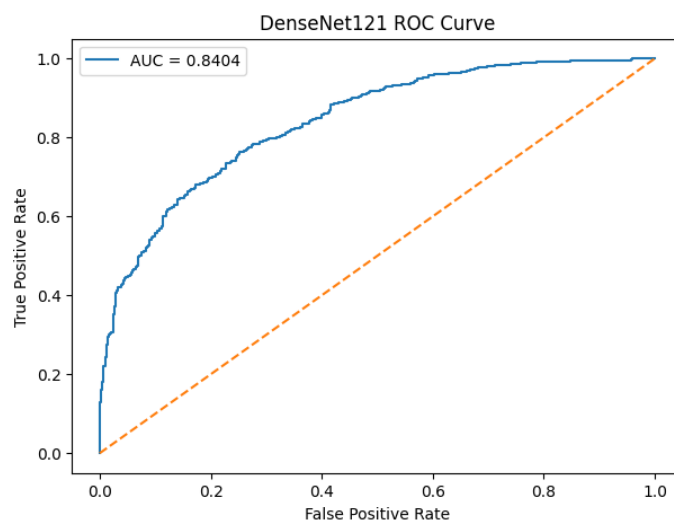


Figure 5.3: ROC Curve of DenseNet121

5.4.2 EfficientNetB0

EfficientNetB0 achieved an accuracy of 78.28%, precision of 83.01%, recall of 83.44%, F1-score of 83.22%, and an AUC of 85.03%. Thanks to its compound scaling method, the model was able to effectively accommodate discriminative features while preserving both computational efficiency and classification accuracy for these histopathological images.

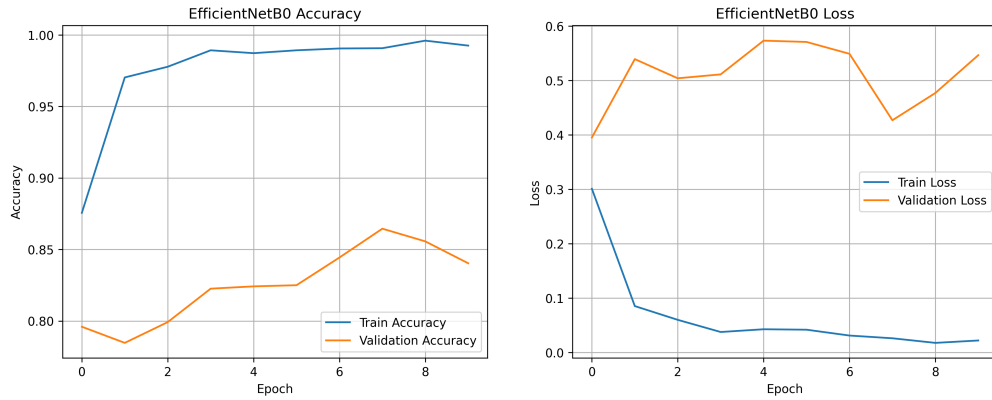


Figure 5.4: Training and Validation Accuracy/Loss Curves of EfficientNetB0

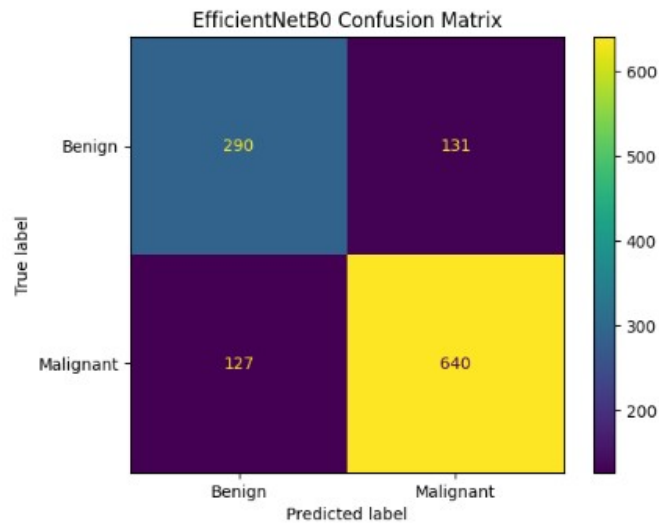


Figure 5.5: Confusion Matrix of EfficientNetB0

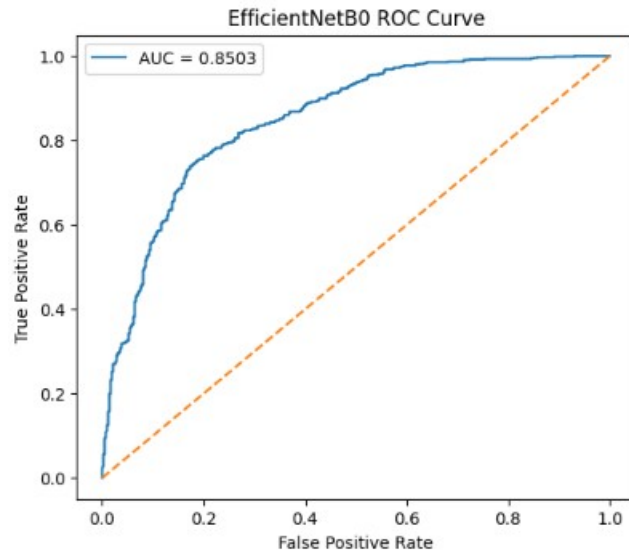


Figure 5.6: ROC Curve of EfficientNetB0

5.4.3 ResNet50

ResNet50 achieved an accuracy of 81.48%, precision of 84.23%, recall of 87.74%, and F1-score of 85.95%. Moreover, it achieved the best AUC value with 88.55%, which has a robust discriminative ability, and a stable classification performance of the models on the BreakHis dataset.

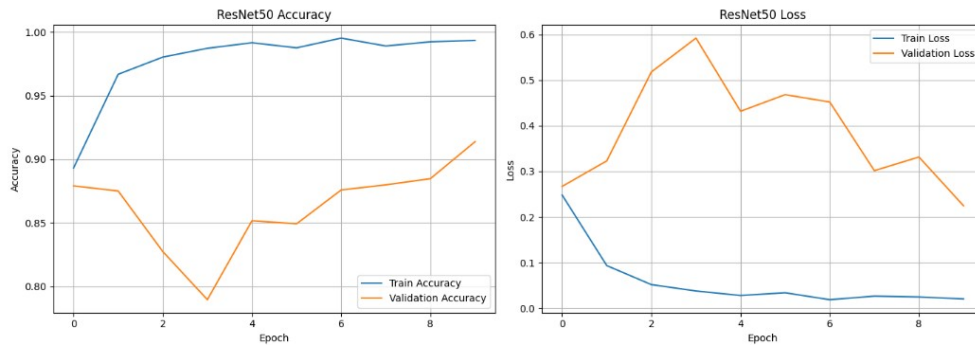


Figure 5.7: Training and Validation Accuracy/Loss Curves of ResNet50

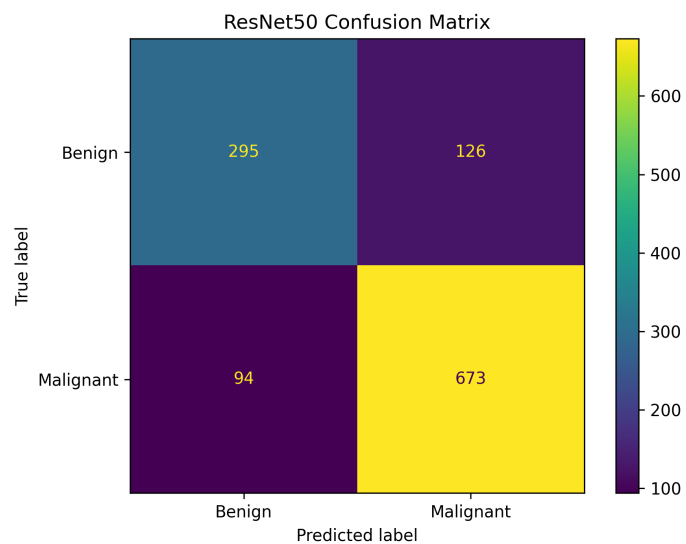


Figure 5.8: Confusion Matrix of ResNet50

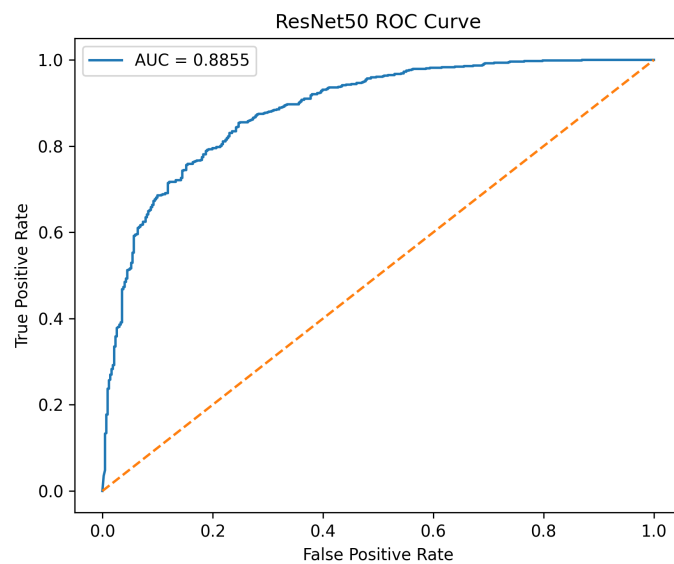


Figure 5.9: ROC Curve of ResNet50

5.4.4 Xception

Xception achieved an accuracy of 79.38%, precision of 81.07%, recall of 88.79%, F1-score of 84.75%, and AUC of 84.46%. The depthwise separable convolution provided the model with excellent feature extraction performance on pathological images, allowing to efficiently learn elaborate patterns of tissues, and also on moderate computation complexity.

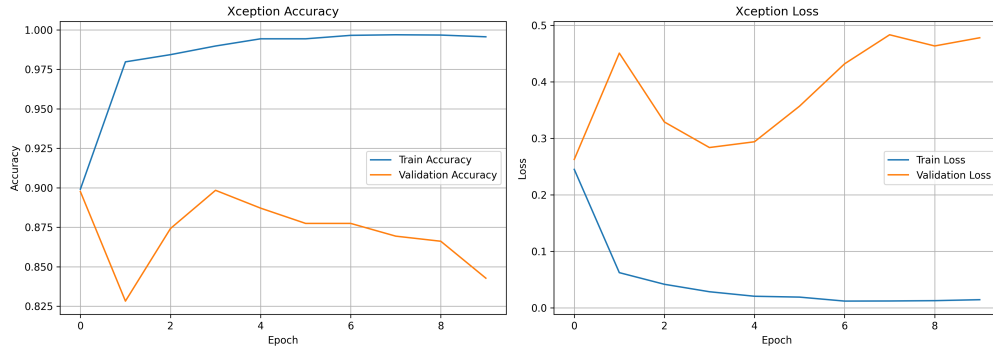


Figure 5.10: Training and Validation Accuracy/Loss Curves of Xception

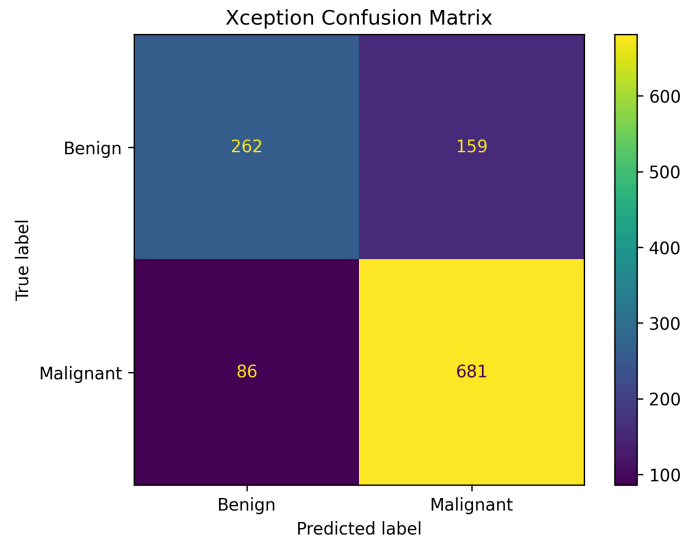


Figure 5.11: Confusion Matrix of Xception

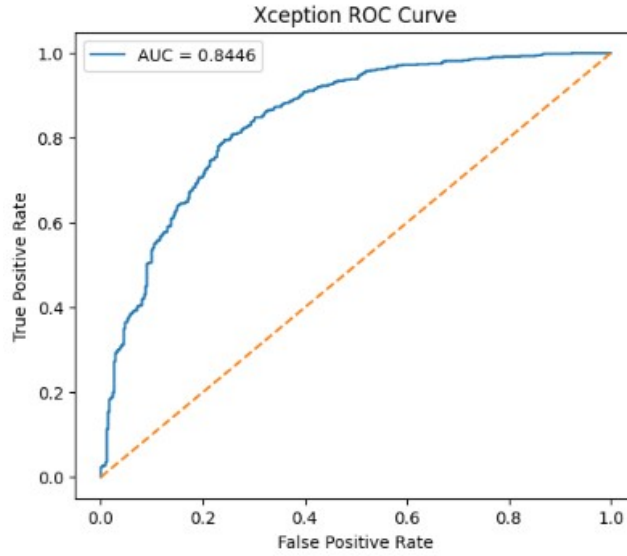


Figure 5.12: ROC Curve of Xception

5.4.5 MobileNetV3

MobileNetV3 achieved the highest accuracy of 81.82%, along with a precision of 84.65%, recall of 87.74%, F1-score of 86.17%, and AUC of 85.00%. The lightweight structure showed competitive classification performance with low computation complexity and it is a potential one for the medical image analysis applications with limited resources.

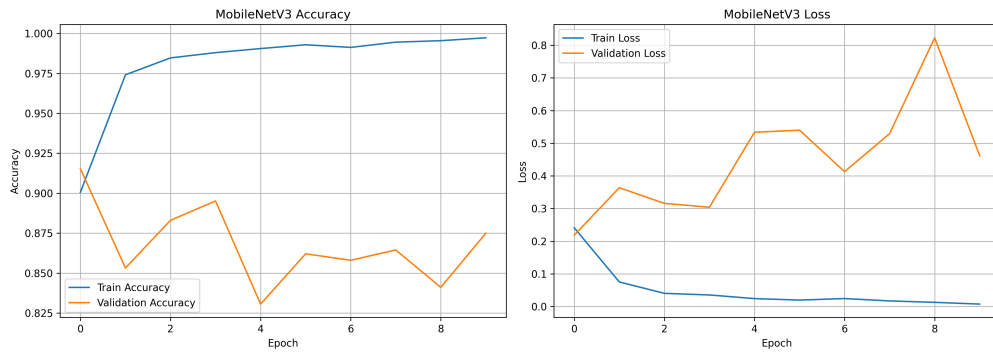


Figure 5.13: Training and Validation Accuracy/Loss Curves of MobileNetV3

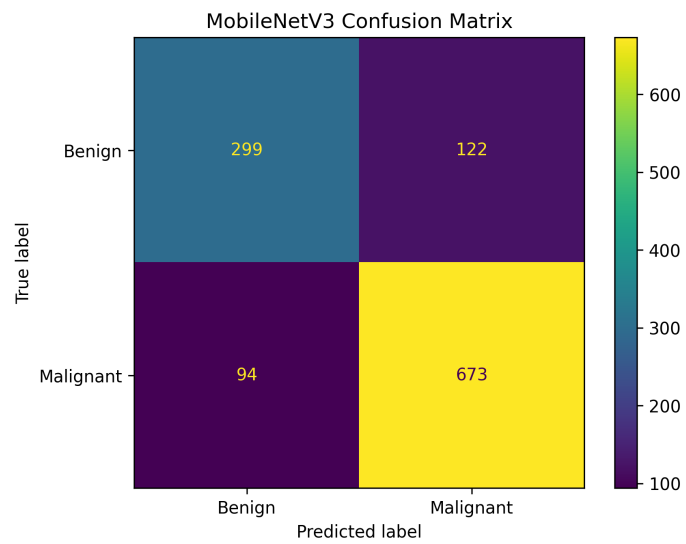


Figure 5.14: Confusion Matrix of MobileNetV3

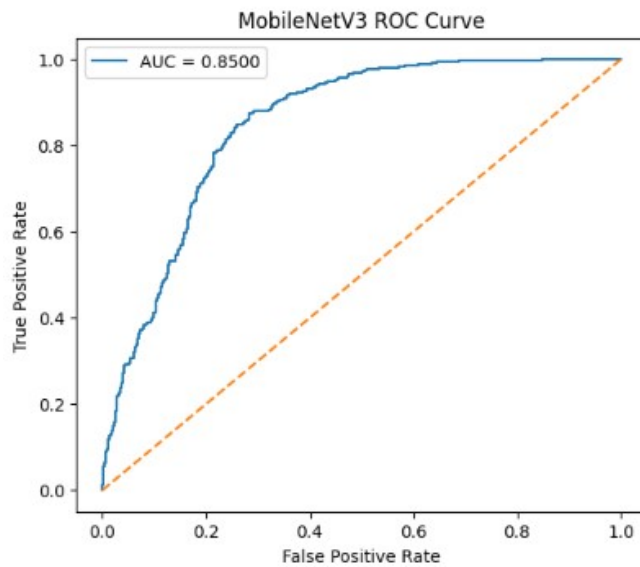


Figure 5.15: ROC Curve of MobileNetV3

5.4.6 Vision Transformer (ViT-B/16)

Vision Transformer (ViT-B/16) achieved an accuracy of 81.57%, precision of 82.70%, recall of 90.35%, F1-score of 86.36%, and AUC of 88.39%. The model achieved the best F1-score and recall compared to the other analyzed models, proving itself effective in extracting global contextual information from histopathological images. The findings in this study show that transformer models can be designed and trained to attain similar levels of performance on BreakHis as comparable network models when they are trained effectively and optimized.

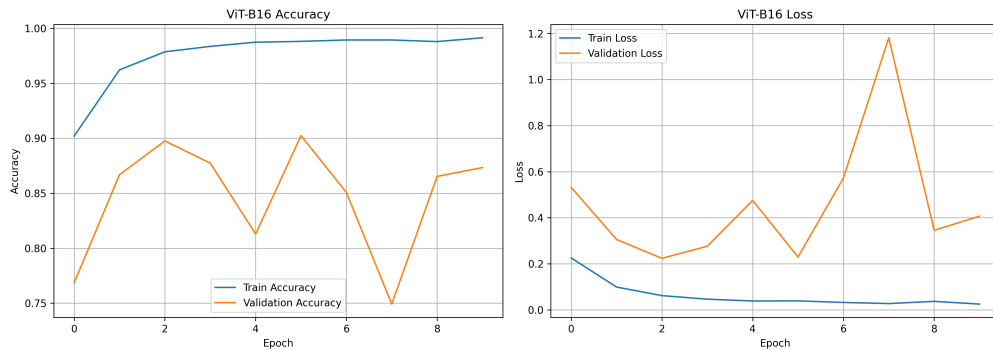


Figure 5.16: Training and Validation Accuracy/Loss Curves of ViT-B/16

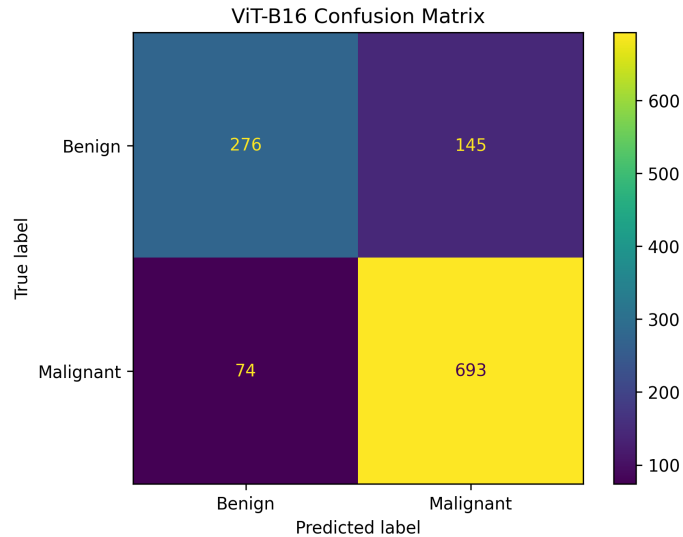


Figure 5.17: Confusion Matrix of ViT-B/16

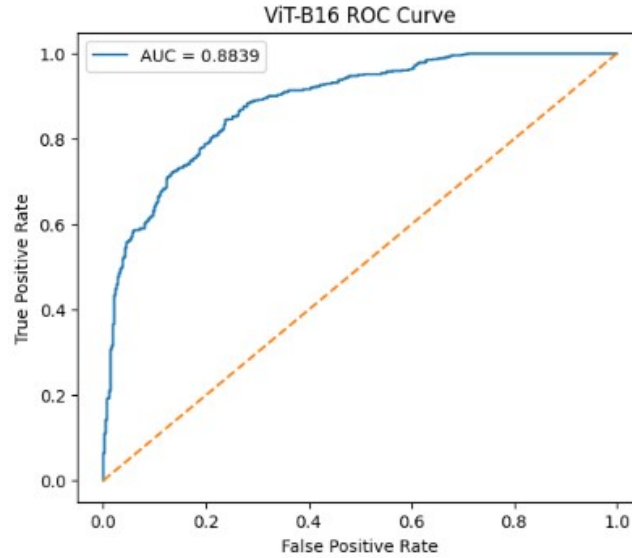


Figure 5.18: ROC Curve of ViT-B/16

5.5 Explainability Results

A model interpretability method is used to create grad-cam++ to improve the interpretability of the best performing model, MobileNetV3. The resulting heatmaps feature identified areas of the histopathological images, such areas with groups of nuclei or with abnormalities to the image, which also play an important role in the ultimate decision on the classification. These graphical explanations make the model more transparent, and make it more credible when predicting real-world results, as relevant to clinical applications.

MobileNetV3 Grad-CAM++

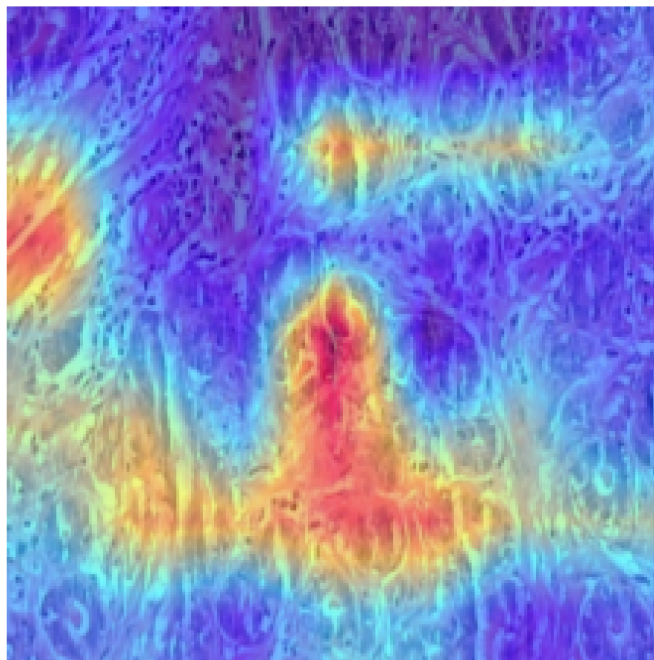


Figure 5.19: MobileNetV3 Grad CAM Plus

6 CONCLUSIONS

This study presented a comprehensive comparative evaluation of six deep learning and transformer-based architectures for breast cancer classification using the BreakHis histopathological image dataset. The evaluated models included DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3, and Vision Transformer (ViT-B/16). Transfer learning and fine-tuning techniques were employed to train the models, and their performance was assessed using Accuracy, Precision, Recall, F1-score, and AUC. The experimental results demonstrated that all models were capable of effectively distinguishing between benign and malignant breast cancer samples. Among the evaluated architectures, ViT-B/16 achieved the highest F1-score (86.36%) and recall (90.35%), indicating strong capability in identifying malignant cases. MobileNetV3 achieved the highest accuracy (81.82%) and precision (84.65%), while ResNet50 attained the highest AUC (88.55%), demonstrating superior discriminative performance across classification thresholds. Xception also showed competitive performance, particularly in capturing fine-grained histopathological features through its depthwise separable convolution architecture. Although DenseNet121 and EfficientNetB0 produced comparatively lower results, they maintained stable classification performance and effective feature representation. Furthermore, Grad-CAM++ was utilized to enhance model interpretability by highlighting diagnostically relevant regions within histopathological images. This explainable AI (XAI) component provides visual insights into the model’s decision-making process, increasing transparency and supporting clinical trust. Overall, the findings indicate that both convolutional neural networks and transformer-based architectures can effectively support automated breast cancer diagnosis from histopathological images. The integration of explainable AI further strengthens the practical applicability of these models in real-world clinical environments, assisting pathologists in achieving earlier and more accurate breast cancer detection.

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